Lab Assignment 3

2O13

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1. Write a program to implement (a) pointer to an object (b) this pointer. Practice both '.' (dot operator) and '→' (arrow operator).

Users > Shagun Garg > oops.cpp > main()

```cpp
#include <iostream>
using namespace std;
class rectangle{
    int length;
    int bredth;

public:
    rectangle(){
        length = 0;
        bredth = 0;
    }
    void setdimensions(){
        cout << "Enter Length: ";
        cin >> length;
        cout << "Enter Breadth: ";
        cin >> bredth;
    }
    void displaydimensions(){
        cout << "Length: " << length << endl;
        cout << "Breadth: " << bredth << endl;
    }
};

int main() {
    rectangle R1;
    rectangle *ptr = &R1;

    R1.setdimensions();
    R1.displaydimensions();
    cout << endl;
    ptr->setdimensions();
    ptr->displaydimensions();
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\User
Enter Length: 4
Enter Breadth: 5
Length: 4
Breadth: 5

Enter Length: 8
Enter Breadth: 9
Length: 8
Breadth: 9
PS C:\Users\Shagun Garg>
```

2. Write a program to swap private values of two classes using a friend function.

```
Users > Shagun Garg > oops.cpp > test1
#include <iostream>
#include <string>
using namespace std;
class test2;
class test1{
    int value1;
public:
    test1(int x){
        value1= x;
    }
    friend void swap(test1 &,test2 &);
    void display() {
        cout << "Value1: " << value1 << endl;
    }
};
class test2{
    int value2;
public:
    test2(int y){
        value2= y;
    }
    friend void swap(test1 &,test2 &);

    void display() {
        cout << "Value2: " << value2 << endl;
    }
};
void swap(test1 &a,test2 &b){
    int temp;
    temp=a.value1;
    a.value1=b.value2;
    b.value2=temp;
}
```

```
int main(){
    test1 t1(10);
    test2 t2(20);
    cout<<"Original Values:"<<endl;
    t1.display();
    t2.display();
    cout<<endl;
    swap(t1,t2);
    cout<<"Swapped values: "<<endl;
    t1.display();
    t2.display();
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\u
Original Values:
Value1: 10
Value2: 20

Swapped values:
Value1: 20
Value2: 10
PS C:\Users\Shagun Garg>
```

3. Write a program to add data objects of two different classes using friend functions.

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
#include <string>
using namespace std;
class test2;
class test1{
    int value1;
public:
    test1(int x){
        value1= x;
    }
    friend void sum(test1 &,test2 &);
    void display() {
        cout << "Value1: " << value1 << endl;
    }
};
class test2{
    int value2;
public:
    test2(int y){
        value2= y;
    }
    friend void sum(test1 &,test2 &);

    void display() {
        cout << "Value2: " << value2 << endl;
    }
};
void sum(test1 &a,test2 &b){
    int s=a.value1+b.value2;
    cout<<s;
}
```

```
int main(){
    test1 t1(10);
    test2 t2(20);
    cout<<"Original Values:"<<endl;
    t1.display();
    t2.display();
    cout<<endl;
    cout<<"Sum of values: "<<endl;
    sum(t1,t2);
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd '
Original Values:
Value1: 10
Value2: 20

Sum of values:
30
PS C:\Users\Shagun Garg>
```

4. Write a program to demonstrate the working of friend class.

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
#include <string>
using namespace std;

class test{
    int a;
    int b;
public:
    test(int x,int y){
        a= x;
        b= y;
    }

    friend int sum(test s);
};
int sum(test s){
   return s.a+s.b;
}
int main(){
    int p,q;
    cout<<"Enter numbers: "<<endl;
    cin>>p;
    cin>>q;

    test t(p,q);
    cout<<"Sum: "<<sum(t);
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd
Enter numbers:
15
8
Sum: 23
PS C:\Users\Shagun Garg>
```

5. Write a program using Array of Objects to display area of multiple rectangles.

```
Users > Shagun Garg > oops.cpp > ...
#include <iostream>
using namespace std;

class Rectangle {
    int length, breadth;
public:
    void setData() {
        cout << "Enter length and breadth: ";
        cin >> length >> breadth;
    }
    int area() {
        return length * breadth;
    }
    void display(int i) {
        cout << "Area of Rectangle " << i + 1 << " = " << area() << endl;
    }
};

int main() {
    int n;
    cout << "Enter number of rectangles: ";
    cin >> n;
    Rectangle r[n];
    for (int i = 0; i < n; i++) {
        cout << "\nRectangle " << i + 1 << endl;
        r[i].setData();
    }
    cout << " Area:\n";
    for (int i = 0; i < n; i++) {
        r[i].display(i);
    }

    return 0;
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PO
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Ga
Enter number of rectangles: 3

Rectangle 1
Enter length and breadth: 4
5

Rectangle 2
Enter length and breadth: 2
8

Rectangle 3
Enter length and breadth: 1
6
 Area:
Area of Rectangle 1 = 20
Area of Rectangle 2 = 16
Area of Rectangle 3 = 6
PS C:\Users\Shagun Garg>
```

6. Write a program to define function ***cube()*** as inline for calculating cube of a number.

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
using namespace std;

inline int cube(int x) {
    return x * x * x;
}

int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;

    cout << "Cube of " << n << " = " << cube(n) << endl;

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "
Enter a number: 6
Cube of 6 = 216
PS C:\Users\Shagun Garg>
```

7. Write a program to pass an object as an argument and return the object from a function.

   a. Use pass-by-value

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
#include <string>
using namespace std;

class student {
    string name;
    int Roll_no;
    float marks;

public:
    void setdetails(string s, int r, float m) {
        name = s;
        Roll_no = r;
        marks = m;
    }

    void displaydetails() {
        cout << "Name: " << name << endl;
        cout << "Roll No.: " << Roll_no << endl;
        cout << "Marks: " << marks << endl;
    }
};

int main() {
    student s1;
    s1.setdetails("Shagun", 23, 89.5);
    s1.displaydetails();
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "
Name: Shagun
Roll No.: 23
Marks: 89.5
PS C:\Users\Shagun Garg>
```

b. Use pass-by reference

```
Users > Shagun Garg > oops.cpp > ...
#include <iostream>
#include <string>
using namespace std;
class student {
    string name;
    int Roll_no;
    float marks;
public:
    void setdetails(string &s, int &r, float &m) {
        name = s;
        Roll_no = r;
        marks = m;
    }
    void displaydetails() {
        cout << "Name: " << name << endl;
        cout << "Roll No.: " << Roll_no << endl;
        cout << "Marks: " << marks << endl;
    }
};
int main() {
    student s1;
    string s;
    int r;
    float m;
    cout<<"Enter name: ";
    cin>>s;
    cout<<"Enter roll no.: ";
    cin>>r;
    cout<<"Enter marks: ";
    cin>>m;
    s1.setdetails(s,r,m);
    s1.displaydetails();
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\U
Enter name: Shagun
Enter roll no.: 1024150104
Enter marks: 89.99
Name: Shagun
Roll No.: 1024150104
Marks: 89.99
PS C:\Users\Shagun Garg>
```